

A Comparative Study on Locomotion Methods and Distance Perception in Immersive Virtual Reality

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ABSTRACT

Accurately perceiving distances in virtual reality (VR) remains a challenge due to discrepancies between real-world and VR spatial experiences. This study compares four VR locomotion methods—Teleport, Joystick, Arms Swinging, and Real Walking—by evaluating distance perception, spatial orientation, and user experience. In an exploratory within-subject study, participants navigated a virtual environment, estimated distances, and completed various questionnaires (SSQ, IPQ, SUS, NASA TLX, and a comparison questionnaire). Results showed a mismatch between user preference and performance: while Arms Swinging was the least preferred method, it provided the most accurate distance estimation and spatial orientation.

Index Terms: Locomotion, virtual reality, distance perception, navigation, spatial awareness.

1 INTRODUCTION

Accurate distance perception is vital for effective interaction in virtual environments, as it enhances spatial awareness and contributes to a heightened sense of presence [7]. Previous studies have mainly focused on display modalities to improve distance perception [2, 7], with less attention given to how locomotion methods affect distance perception. Although much of the existing literature evaluates locomotion methods based on usability, presence, and spatial knowledge [1, 3, 4], their influence on distance perception remains largely unexplored. While early framework for egocentric distance perception suggested that perceived distance might be independent of locomotion [6], a more nuanced understanding is needed, particularly in the context of modern head-mounted displays and current technology [5].

This study aims to address this gap by evaluating the impact of four common VR locomotion methods—Teleport, Joystick, Arms Swinging, and Real Walking—on distance perception. While teleportation is often used as a control in VR studies [1], we use Real Walking as our control condition for a more realistic comparison. Through an exploratory within-subjects user study, participants moved between various locations in a virtual environment, estimated travelled distances, and performed a spatial awareness task to locate their starting position.

2 METHODOLOGY

The user study was carried out after approval from the university ethics research committee of the University of Genoa (n. 2024/45).

Study Participants: The user study involved 28 volunteers (male = 20; female = 8; age = 27.85 ± 2.96). The sample size was calculated based on a power analysis performed on G*Power v3.1 (repeated measures ANOVA; four conditions; significance level =

0.05; effect size = 0.30; statistical power = 0.95). All participants were healthy and had normal or corrected-to-normal vision. Before participation, they gave their informed consent.

Task: Participants navigated to various color-coded waypoints in the virtual environment and estimated the perceived distances travelled (see Figure 1 for an overview). Starting from the center of the room, they first located and moved to a yellow disc, which marked the task's starting point. Upon reaching the yellow disc, it disappeared and a red disc appeared. The participants moved to the red disc, after which a GUI prompted them to estimate the distance travelled. This process repeated for blue and purple discs. Upon reaching the purple disc, participants performed an additional task: using a raycast, they pointed to the perceived location of the initial yellow disc. Each session included five iterations, requiring participants to estimate 15 distances and locate the starting position five times. Disc positions were randomized, forming a trapezoidal pattern, with distances between waypoints ranging from 2 to 7 meters.

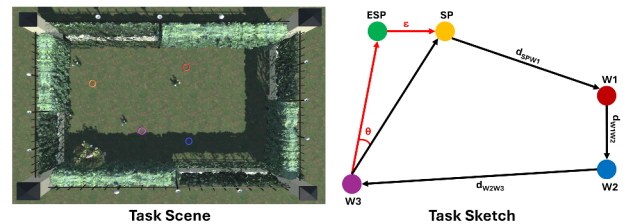


Figure 1: (left) The task scene where color-coded rings represent the waypoints. (right) Illustration of the user task. SP: starting position; W1: waypoint 1; W2: waypoint 2; W3: waypoint 3; ESP: estimated starting position; d_{SPW1} : distance between SP and W1; d_{W1W2} : distance between W1 and W2; d_{W2W3} : distance between W2 and W3; θ : angle error; ϵ : position error.

Locomotion Conditions: Participants experienced four locomotion methods: Real Walking, Teleport, Joystick, and Arms Swinging. In Real Walking, participants physically moved through waypoints, with their real-world movements mirrored in the virtual environment. In Teleport, participants selected a target location by pointing and pressing a trigger on the Vive controller to instantly move to that position. Joystick locomotion allowed users to navigate by utilizing the trackpad on the Vive controller, where the thumb's position on the trackpad dictated the direction and speed within the virtual environment. For Arms Swinging, participants simulated walking by swinging their arms alternately (e.g., left arm up as right arm goes down). The system interpreted this motion as forward movement, with speed increasing proportionally to the arm-swinging intensity.

Procedure: The experiment was conducted in an $8m \times 4.5m$ empty room with four Vive base stations to ensure full tracking coverage. Participants wore an HTC Vive Pro Eye headset, with their interpupillary distance (IPD) measured and adjusted. Before each session, participants completed a practice session in a virtual environment to familiarize themselves with the assigned locomotion method. Once comfortable, they proceeded to the main task using

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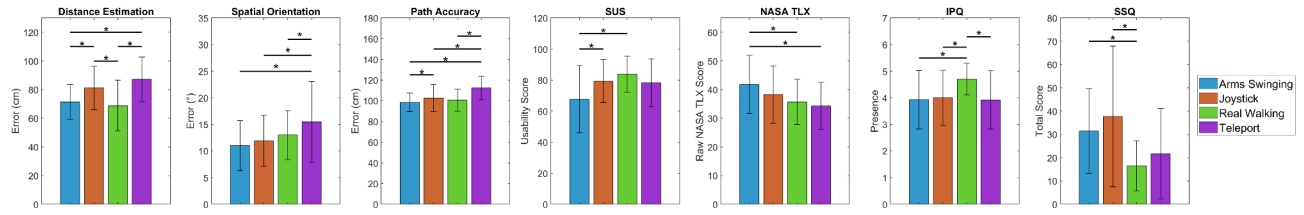


Figure 2: Computed user metrics. Errors in distance estimation, spatial orientation and path accuracy along with user experience questionnaires are reported. Statistically significant differences ($p < 0.05$) between locomotion methods are indicated with *.

one of the locomotion methods. After each session, participants completed user experience questionnaires (SSQ, IPQ, NASA TLX, SUS). This process was repeated for all four locomotion methods, with the session order randomized for each participant. After completing all four locomotion sessions, participants completed a comparison questionnaire.

3 RESULTS

The recorded data was analyzed to quantify user performance. To understand statistical differences between different locomotion conditions based on user metrics, repeated measures ANOVA test was performed. The computed user metrics are shown in Figure 2.

Estimated Distances: Analysis of signed errors revealed that approximately half of the participants overestimated distances while the other half underestimated them. To better assess accuracy, absolute errors were analyzed, revealing significant differences. Real Walking and Arms Swinging exhibited comparable performance with significantly lower errors than Joystick and Teleport, indicating that Arms Swinging matched Real Walking in distance estimation accuracy and outperformed the other methods.

Spatial Orientation: Spatial orientation was evaluated by calculating the angle error, θ , between the vectors $\overrightarrow{W3SP}$ and $\overrightarrow{W3ESP}$. For angle error, Arms Swinging, Joystick, and Real Walking demonstrated statistically similar performance and were significantly better than Teleport. This suggests that participants were more accurate in estimating their starting positions using Arms Swinging, Joystick, and Real Walking compared to Teleport.

Path Accuracy: The users' recorded trajectories were analyzed against their distance estimations, factoring in cumulative distance and path curvature. Results indicated that Teleport was significantly less accurate compared to the other three methods, which demonstrated similar accuracies. Additionally, trajectory analysis revealed that users required frequent trajectory corrections in Arms Swinging, as they faced challenges in maintaining a straight path.

User Experience: Statistical analysis of SUS revealed that the Arms Swinging method had significantly lower usability scores compared to Joystick and Real Walking, while no significant differences were found among the other methods. Simulator sickness was lowest with Real Walking, likely due to reduced sensory conflict from physical movement aligning visual and vestibular cues. In contrast, Joystick induced the highest sickness levels, likely from sensory mismatch, while overall sickness severity was comparable between Real Walking and Teleport. General presence were significantly higher with Real Walking than with other methods, suggesting enhanced immersion and realism through physical movement. Arms Swinging had the highest perceived workload, significantly more than Real Walking and Teleport.

User Feedback: Real Walking was the most preferred locomotion method, while Arms Swinging was the least favored (Arms Swinging = 4.35; Joystick = 6.39; Real Walking = 8.43; Teleport = 5.83). Interestingly, this contrasts with performance, where Arms Swinging showed better accuracy in distance judgments and spatial awareness. Participants reported difficulty judging starting po-

sitions in the Teleport condition, while Arms Swinging was considered physically tiring, with some participants experiencing mild dizziness. In Real Walking, a few noted a fear of falling, possibly due to the absence of a user avatar representing their limbs. Nausea was occasionally reported in the Joystick session.

4 CONCLUSION

This study explored the impact of different VR locomotion methods—Teleport, Joystick, Arms Swinging, and Real Walking—on distance perception. In an exploratory within-subjects user study, participants navigated a virtual environment, estimated distances, and identified their starting position. Results showed a clear preference for Real Walking, while Arms Swinging, despite being less favored, led to better task performance. As expected, simulator sickness was highest with Joystick, and physical load was greater with Real Walking and Arms Swinging. Interestingly, there was no significant difference in presence between Teleport, Joystick, and Arms Swinging. These findings highlight the impact of locomotion methods on VR experiences. Future work will address embodiment issues in VR locomotion by incorporating user avatars.

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